27. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{(n+1)(2n+1)}$.

PART – C (5 × 12 = 60 Marks)
Answer **ALL** Questions

28. a. Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix}$ and hence find A^{-1}.

(OR)

b. Reduce the quadratic form $6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$ to canonical form by an orthogonal transformation. Also state its nature.

29. a. Explain $e^x \cos y$ in powers of x and y as far as the terms of the third degree.

(OR)

b. Find the volume of the largest rectangular parallelepiped that can be inserted in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

30. a. Solve $(D^2 - 7D + 12)y = e^{5x} + \cos 2x$.

(OR)

b. Solve $(x^2D^2 + xD - 9)y = \frac{5}{x^2}$.

31. a. Find the equation of the circle of curvature of the parabola $y^2 = 12x$ at the point (3, 6).

(OR)

b. Find the evolute of the parabola $y^2 = 4ax$.

32. a.i. State the axioms of convergence of series by comparison test.

ii. Test the convergence of the series $\frac{3}{4} + \frac{3.4}{4.6} + \frac{3.4.5}{4.6.8} + \ldots\infty$.

(OR)

b.i. State the Cauchy's criterion of convergence of an infinite series.

ii. Test the convergence of the series $\sum_{n=1}^{\infty} \left(\sqrt{n^4+1} - \sqrt{n^4-1} \right)$.

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Reg. No.															

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018
First Semester

18MAB101T – CALCULUS AND LINEAR ALGEBRA
(For the candidates admitted during the academic year 2018 – 2019)

Note:
(i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)
Answer **ALL** Questions

1. The eigen values of A^2, given $A = \begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$ are
(A) 3, 2, 5
(B) 9, 4, 25
(C) 3, 1, 4
(D) 9, 1, 16

2. The characteristics equation of the matrix $A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$ is
(A) $\lambda^2 - 4\lambda - 5 = 0$
(B) $\lambda^2 + 4\lambda - 5 = 0$
(C) $\lambda^2 + 4\lambda + 5 = 0$
(D) $\lambda^2 + 4\lambda - 5 = 0$

3. The nature of the quadratic form, whose matrix is $\begin{pmatrix} 1 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 6 \end{pmatrix}$ is
(A) Indefinite
(B) Positive semi-definite
(C) Positive definite
(D) Negative definite

4. If the canonical form of a quadratic form is $-y_1^2 + y_2^2 + 4y_3^2$, then the signature of QF is
(A) 0
(B) 3
(C) 2
(D) 1

5. If $u = xy^2 + x^2y$, where $x = at^2$, $y = 2at$, that $\frac{du}{dt}$
(A) $2a^2t^2(8+5t)$
(B) $2a^3t^3(8+5t)$
(C) $4a^3t^3(8+5t)$
(D) $4a^2t^2(8+5t)$

6. The stationary points of the function $3x^2 - y^2 + x^3$ are
(A) (0, 0), (–2, 0)
(B) (0, 0), (2, 0)
(C) (1, 1), (–2, 0)
(D) (1, 1) (2, 0)

7. The $x = r\cos\theta, y = r\sin\theta$, then the Jacobian of x, y with respect to r, θ $\left(i.e\ J\left(\frac{x,y}{r,\theta}\right)\right)$

(A) r (B) $1/r$
(C) $2r$ (D) $2/r$

8. Maclaurin series for e^{x+y} is

(A) $1+(x+y)-\frac{(x+y)^3}{3!}+....$ (B) $1-(x-y)-\frac{(x-y)^3}{2!}-....$
(C) $1+(x-y)-\frac{(x-y)^3}{2!}+....$ (D) $1+(x+y)+\frac{(x+y)^3}{2!}+....$

9. The solution of $(D^2+4)y = 0$ is

(A) $y = Ae^{2x} + Be^{2x}$ (B) $y = A\cos 2x + B\sin 2x$
(C) $y = Ae^{2x} + Be^{-2x}$ (D) $y = A\cos x + B\sin x$

10. The particular integral of the differential equation $(D^2 - 3D + 2)y$ cos$x$ is

(A) $\frac{1}{10}(\cos x + 3\sin x)$ (B) $\frac{1}{9}(\cos x - 3\sin x)$
(C) $\frac{1}{10}(\cos x - 3\sin x)$ (D) $\cos x - 3\sin x$

11. The complementary function of $(D^2 + 4D + 4)y = e^{-2x}$ is

(A) $CF = (Ax + B)e^{-2x}$ (B) $CF = (Ax + B)e^{2x}$
(C) $CF = A\cos 2x + B\sin 2x$ (D) $CF = Ae^{-2x} + Be^{-2x}$

12. The complementary function of $\left(x^2D^2 + 4xD + 2\right)y = x^2 + \frac{1}{x^2}$ is

(A) $CF = (Ax + B)e^{-2x}$ (B) $CF = (Ax + B)e^{-x}$
(C) $CF = Ae^{-x} + Be^{-2x}$ (D) $CF = \frac{A}{x} + \frac{B}{x^2}$

13. The radius of curvature of the curve $y = e^x$ at $x = 0$ is

(A) $2\sqrt{2}$ (B) $\sqrt{2}$
(C) 2 (D) 4

14. The radius of curvature of a circle at any point is same as its

(A) Chord (B) Radius
(C) Diameter (D) Tangent

15. The value of $\Gamma\left(\frac{1}{2}\right)$ is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$
(C) π (D) $\sqrt{\pi}$

16. When n is a +ve integer, $\Gamma(n+1) =$

(A) $(n+1)!$ (B) $n!$
(C) $(2n)!$ (D) $(n-1)!$

17. The series $\sum_{n=1}^{\infty}\frac{1}{n^r}$ is convergent, if

(A) p > 1 (B) p < 1
(C) p = 1 (D) p = 0

18. The series $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+....$ is

(A) Convergent (B) Oscillating
(C) Divergent (D) Monotonic

19. As per D'Alembert's ratio test, if $\sum u_n$ is a series of positive terms and $\lim_{n\to\infty}\frac{u_{n+1}}{u_n} = l$, the series is convergent, if

(A) $l > 1$ (B) $l = 0$
(C) $l = 1$ (D) $l < 1$

20. The series $\sum\frac{1}{n}\sin\left(\frac{1}{x}\right)$ is

(A) Convergent (B) Conditionally convergent
(C) Divergent (D) Absolutely convergent

PART – B (5 × 4 = 20 Marks)
Answer **ANY FIVE** Questions

21. Use Cayley-Hamilton theorem to find A^3, given that $A = \begin{pmatrix} -1 & 3 \\ 2 & 4 \end{pmatrix}$.

22. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$

23. Find the Jacobian of y_1, y_2, y_3 with respect to x_1, x_2, x_3 if $y_1 = \frac{x_2x_3}{x_1}, y_2 = \frac{x_3x_1}{x_2}$ *and* $y_3 = \frac{x_1x_2}{x_3}$.

24. Find the eigenvalues of the matrix $A = \begin{pmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}$.

25. Solve the equation $(D^2 - 4D + 3)y = \sin 3x$.

26. Find the envelope of the family of straight lines given by $x\cos\alpha + y\sin\alpha = a\sec\alpha$.